Observability Lab: Enterprise SLO Reliability Engineering & PromQL Dashboard

Duration : 90 Min.

Scenario

An SRE team needs a production-grade Grafana dashboard tracking enterprise SLOs, Error Budgets, and Golden Signals using PromQL.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Write advanced PromQL queries, configure Grafana alert rules, design SLO burn rate dashboards, and author reliability engineering guidelines.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. Constructing advanced PromQL queries for 99th percentile latency and error rates: 3 marks
2. Building Grafana SLO dashboard panels visualizing Error Budget consumption: 3 marks
3. Configuring multi-window multi-burn-rate alert rules in Prometheus: 3 marks
4. Designing Golden Signal dashboard panels with dynamic variable dropdowns: 3 marks
5. Formulating Error Budget policy enforcement rules for engineering teams: 2 marks
6. Documenting runbook procedures for responding to rapid budget burn alerts: 2 marks
7. Establishing telemetry retention and aggregation rules to optimize Prometheus storage: 2 marks
8. Authoring a complete Site Reliability Engineering (SRE) operational handbook: 2 marks

Total Score: 20 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.